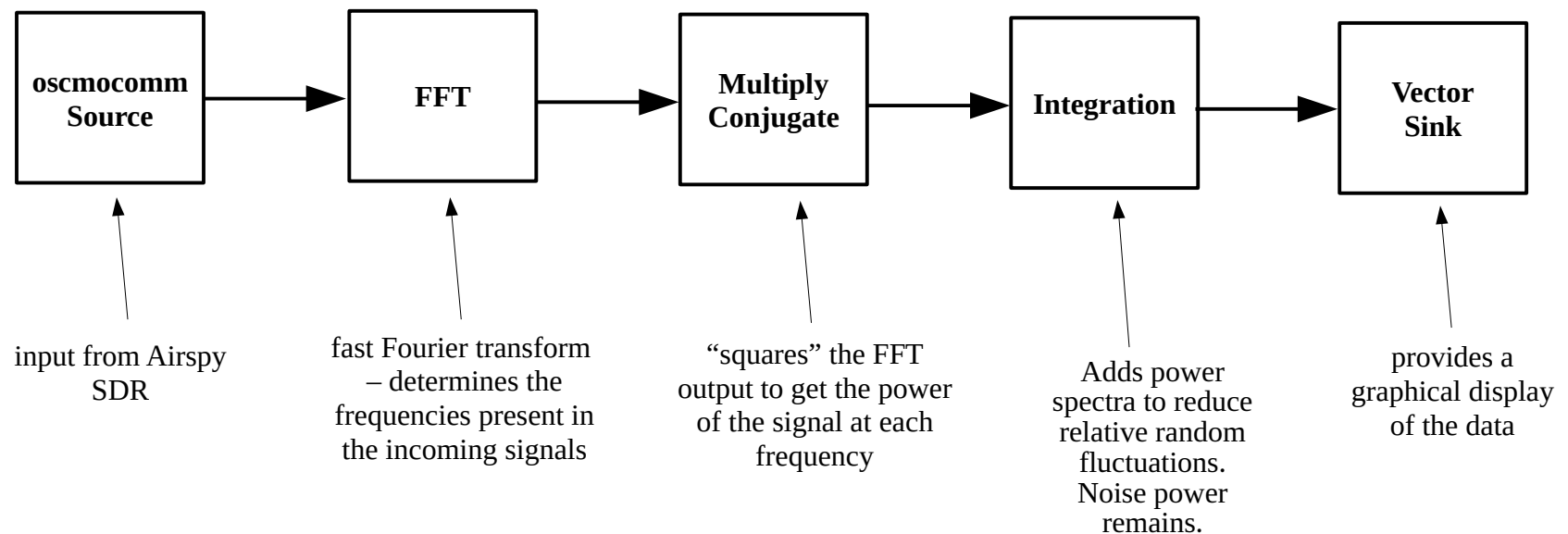


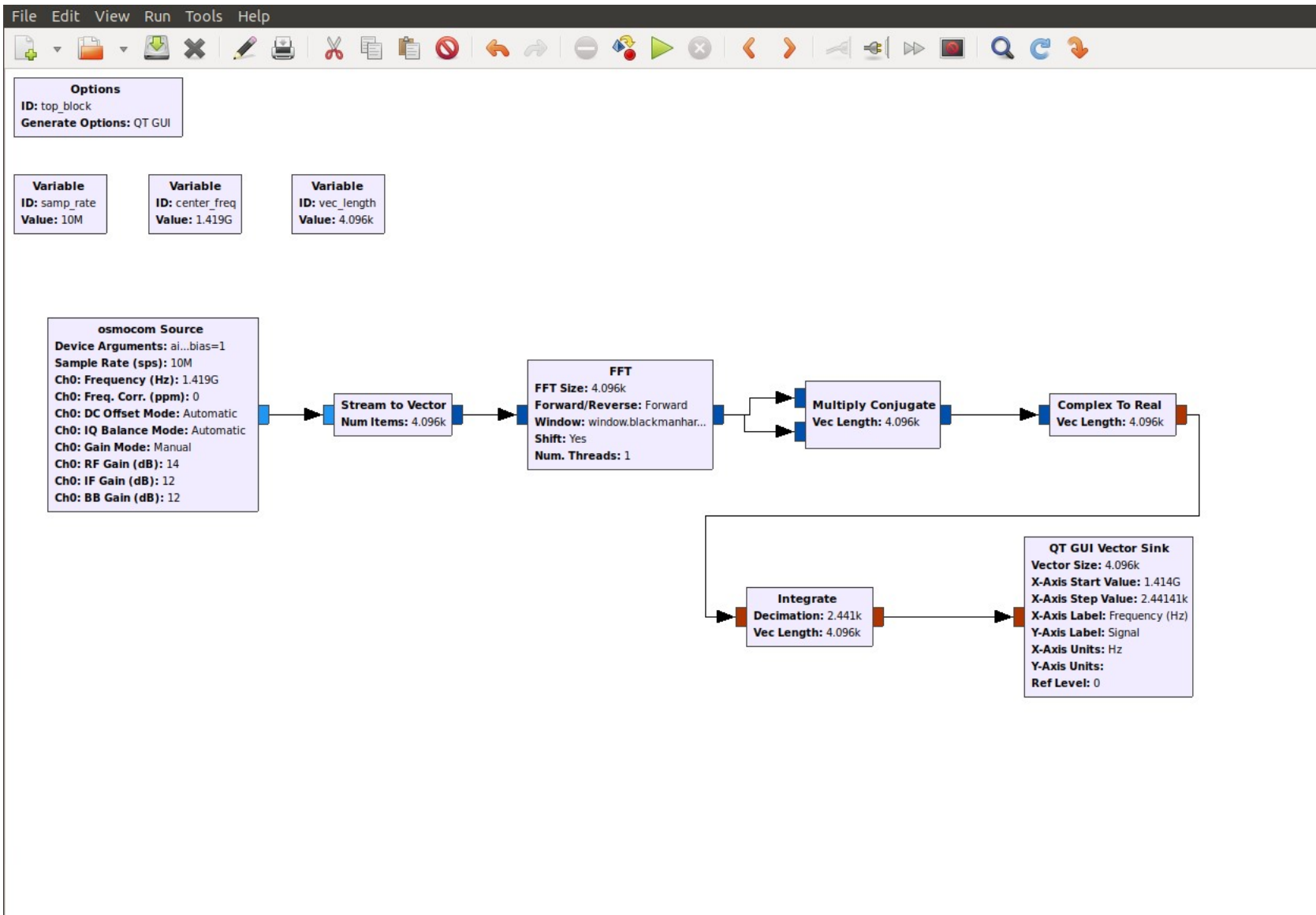
Building a GnuRadio Spectrometer for an HI Radio Telescope Explained

Part 1: A Simple Spectrometer

Block diagram, in its simplest form:



The completed Gnuradio program:



Building the Spectrometer, with Explanations

osmocomm Source block:

- indicates the Airspy as the input block, and its settings.

Settings:

- Device Arguments
 - airspy=0 → identify Airspy as the input device
 - bias=1 → enable antenna-cable power for a compatible LNA
- sample Rate: `samp_rate` → value set as 10 MHz in **Variable** block
- Ch0: Frequency: `center_freq`
 - indicates middle frequency of the 10 MHz Airspy bandwidth, which we will set at 1419 MHz in the `center_freq` **Variable** block.
- RF Gain: 17
- IF Gain: 12
- Leave the other settings at their default values.

USB powers the Airspy itself.

Properties: osmocom Source

General Advanced Documentation

ID: osmosdr_source_0

Output Type: Complex float32

Device Arguments: airspy=0,bias=1

Sync: don't sync

Num Mboards: 1

Mb0: Clock Source: Default

Mb0: Time Source: Default

Num Channels: 1

Sample Rate (sps): samp_rate

Ch0: Frequency (Hz): center_freq

Ch0: Freq. Corr. (ppm): 0

Ch0: DC Offset Mode: Automatic

Ch0: IQ Balance Mode: Automatic

Ch0: Gain Mode: Manual

Ch0: RF Gain (dB): 17

Ch0: IF Gain (dB): 12

Ch0: BB Gain (dB): 12

Ch0: Antenna:

Ch0: Bandwidth (Hz): 0

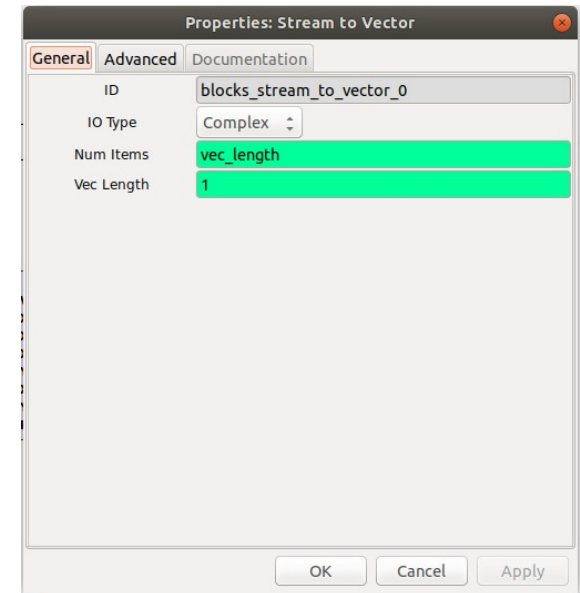
OK Cancel Apply

Stream to Vector block:

- bundles the continuous data stream from the **osmocomm** block into vector chunks to be processed by the FFT block.

Settings:

- **Num items:** `vec_length` → variable set by a **Variable** block (value = 4096)
- **Vec Length:** 1
- NOTE: This nomenclature is confusing; the **Vec Length** is not really a vector length; it indicates that 1 bunch of 4096 **Num Items** is sent into the FFT at a time. This is not to be confused with the variable `vec_length`, which does describe a vector length.

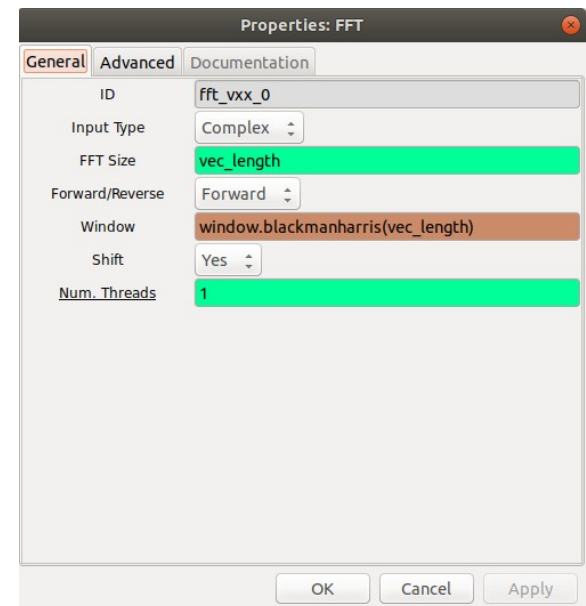


FFT block:

- performs a Fast Fourier Transform on the data, one vector stream at a time.

Settings:

- Input Type: Complex
- FFT Size: `vec_length` (value = 4096)
- Window: `window.blackmanharris(vec_length)`
- NOTE: There is a vector size needed in the **Window** function that must match the FFT size. (It's easy to overlook this.)

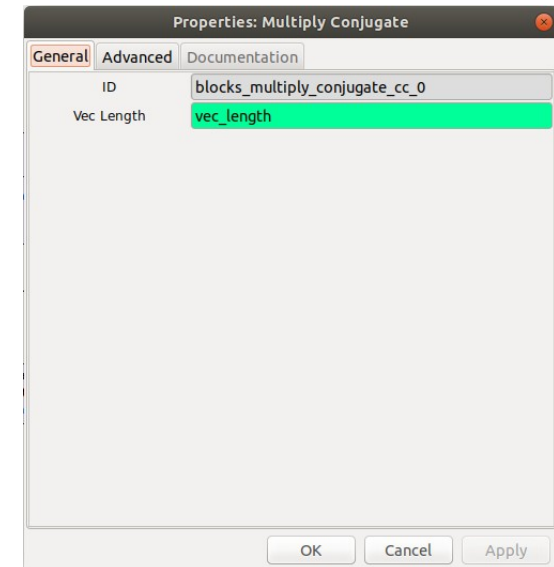


Multiply Conjugate block:

- calculates the product of the FFT output with its complex conjugate.
- This results in a computation of the spectrum power, which is what we want displayed.

Settings:

- Vec Length: vec_length

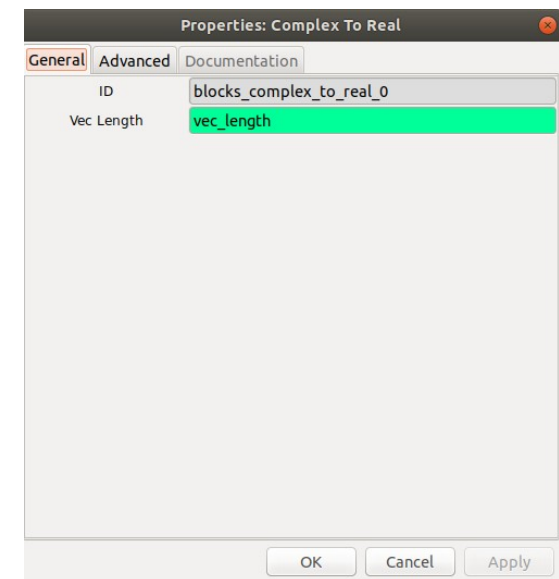


Complex to Real:

- Converts the data stream to a real value.

Settings:

- Vec Length: vec_length



Integrate:

- Adds corresponding power channels across successive spectra. Random fluctuations decrease relative to the mean as more spectra are combined. The mean noise power remains; it does not cancel.

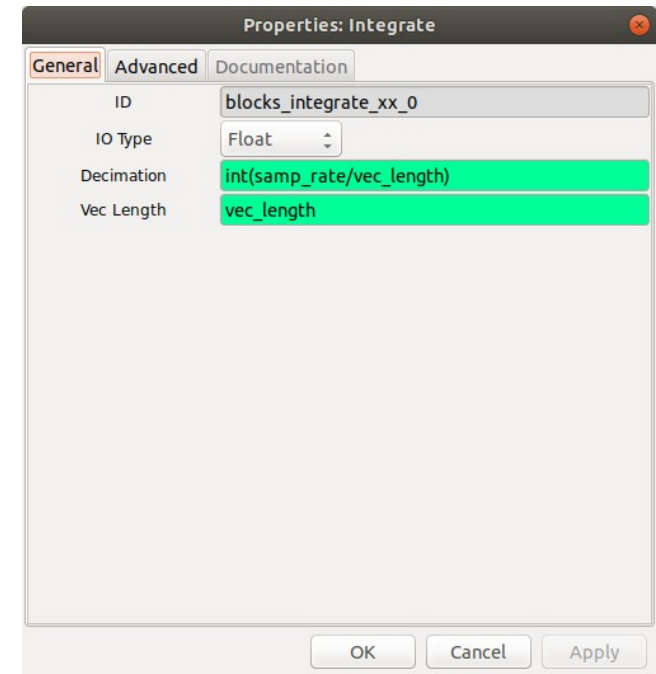
Settings:

- IO Type: Float
- Decimation: `int(samp_rate/vec_length)`
 - Number of complete spectra to add for each output vector.
 - Example:

At 10 MHz and 4096 samples per FFT,
 $10,000,000 / 4096 = 2441.40625$ spectra per second.

Decimation 2441 adds 2441 spectra, giving about one output per second. Divide by 2441 if you want an average. Summing also increases the mean noise level.

- Vec Length: `vec_length`



QT GUI Vector Sink:

- This displays the processed signal in a signal vs. frequency graph.

Settings:

- Vec Size: `vec_length`
- X-Axis Start Value: 1414 MHz = 1414e6
 - We want the 1420.4 MHz of the HI signal to be displayed approximately in the middle of the graph. We will set the 10 MHz sample rate of the Airspy to start at 1414 MHz and stop at 1424 MHz.
- X-Axis Step value: `samp_rate/vec_length`
 - $10,000,000 \text{ Hz} / 4096 = \text{about } 2441 \text{ Hz per channel}$.
There are 4096 channels across the 10 MHz bandwidth.
- Axis labels are as indicated.
- NOTE: The axis units are not automatically displayed on the graph, which is why they are included in the **X-Axis Label**. The **X-Axis Units** setting will enable the units of the x-axis (Hz) to be displayed when a the cursor is moved to an (x, y) coordinate on the screen.
- Autoscale: Yes
- Y min: 0
- Ymax: whatever (since Autoscale is on)
- Leave other settings at their default values.

